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Project Title:	NSCOR: The contribution of non-targeted effects in HZE cancer risk		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	NOTE: Former affiliation was Lawrence Berkeley National Laboratory; moved to NYU in August 2008. Moved to UCSF September, 2015		
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Task Description:	Clear mechanistic understanding of the biological processes elicited by high LET radiation increase cancer risk will reduce the uncertainties in assessing the risk of space flight. Classic targeted radiation effects like cell kill and mutation are proportional to dose and their LET dependence is described by relative biological effectiveness. However recent data suggests that high LET radiation affects multicellular processes that promote cancer. We propose that radiation carcinogenesis results from two classes of ionizing radiation effect: targeted effects that alter genomic sequence and non-targeted effects (NTE) that alter phenotype and multicellular interactions. Since NTE are not proportional to dose, new approaches are required to assess the impact of NTE, which alter gene expression, cell signaling, genomic stability, and cell phenotypes, on carcinogenesis. It is unknown to what extent NTE contribute to the significantly greater carcinogenic effect of high LET radiation. We hypothesize that NTE promotes epithelial cancer. Project 1 will test use novel transgenic and humanized mouse mammary models to evaluate the relative contribution of radiation targeted and NTE to carcinogenesis as a function of radiation quality and of age a radiation chimera model. We will monitor inflammation as a key process affected by radiation NTE and conduct a systems biology analysis of mammary gland. Project 2 will employ new systems genetic analysis of cancer susceptibility in radiation sensitive organs to define critical regulatory networks that impact the relative carcinogenic effectiveness of high LET radiation. Project 3 will use Monte Carlo simulations of targeted radiation effects and agent based modeling of cell interactions to model the relative contributions of targeted and NTE to carcinogenesis. This NSCOR will provide important data on radiation quality dependence of targeted and non-targeted radiation effects and the genetic determinants of cancer susceptibility, which together will used to reduce uncertainties in NASA radiation risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	First, radiation can be considered an experimental probe that perturbs tissue function. Studying these perturbations is a means to define critical biological response to stress. Second, we will develop a comprehensive database of normal tissue and irradiated tissue by analyzing different organs in each mouse. This database will become a public resource. Third, our studies will identify genetic predisposition to radiation associated cancer, and may thus help protect humans who have the greatest risk of cancer.
	Our NSCOR provides experimental studies and modeling to define the efficiency and physiological context in which high LET radiation increases epithelial cancer risk. We hypothesize that both targeted and NTE contribute to radiation carcinogenesis. Targeted effects of ionizing radiation result from interaction of energy with DNA, leading to DNA damage responses and changes in genomic sequence. On the other hand, poorly understood NTE alter phenotype and multicellular interactions that could contribute to cancer. We hypothesize that the greater carcinogenic risk from high LET irradiation is due to NTE promotion of targeted radiation effects. The data generated by the NYU/LBNL/UCSF NSCOR will provide a comprehensive analysis of LET dependence for solid tumor frequency in different tissues, integrated with computational modeling and systems biology to identify key events. The goal of Project 1 is to determine the contribution of NTE as a function of radiation quality on breast cancer. Aim 1.1 uses a mammary chimera model in which wildtype Balb/c mice are transplanted orthotopically with syngeneic Trp53 null mammary epithelium. Tumorigenesis was compared in mice irradiated with different particles before transplantation (i.e. the radiation chimera, in which epithelium is not exposed), irradiation of intact mammary glands (i.e. the genetic chimera, in which a Trp53 null epithelial outgrowth is irradiated in situ), and in aged mice in which 10 month old mice were irradiated before transplantation. Aim 1.2 investigates the inflammatory signals induced by radiation. Aim 1.3 uses systems biology to integrate data from different models and types to provide a better mechanistic understanding of underlying biology. It is unknown to what extent the carcinogenic potential as a function of radiation quality is mediated by differential effects on the irradiated microenvironment. To address this issue, we employed a radiation chimera mammary model to separate radiation-induced microenvironment from radiation-induced mammary epithelial DNA damage and potential mutation. Our current analyses show that NTE do contribute to carcinogenesis by affecting tumor latency, growth rate, and characteristics. Tumors affected only by radiation NTE, i.e. arising in irradiated mice, were far more aggressive than those from non-irradiated mice. In contrast tumors arising from irradiated Trp53 epithelium were distinct in that neither ER-negative tumors nor metastases were increases. We conclude that aggressive tumors are promoted by radiation NTE host responses such as oscillatory inflammation. We have submitted a manuscript that shows that Trp53 tumors arising in Si-irradiated hosts are enriched in a K18-positive, ER-negative subtype but these tumors are different from similar tumors that arose in sham-irradiated mice. These analyses show that NTE do contribute to carcinogenesis by affecting tumor latency, tumor growth rate, and tumor features. The biological behavior and features of tumors arising from un-irradiated Trp53 null mammary epithelium transplanted to sham-irradiated mice were compared to tumors arising in mice contemporaneously irradiated with 350 MeV/amu Si particles or a reference dose of sparsely ionizing gamma-radiation. Tumors arising in mice exposed to densely ionizing radiation grew significantly faster, were more likely to metastasize, and could be genomically separated from tumors with the same markers arising in sham-irradiated mice. These data suggest that the greater carcinogenic effect of densely ionizing radiation is mediated in part by the microenvironment, which elicits more aggressive tumors compared to similar tumors arising in sham-irradiated hosts. Although we do not observe a robust LET and fluence dependence, given our evolving understanding of the NTE dose-dependence, it may well be that thresholds rather than proportional responses may be more critical. For example, it may be that each NTE has a distinct threshold and that these change as a function of LET. The effect on tumor type (e.g. ER statues) appears to be the most sensitive index in the radiation chimera model. A dose threshold for affecting tumor ER status was exceeded by 10 cGy low LET radiation or with a fluence of 1 particle/100 microm2 of high LET particles. The lack of classical dose dependence is challenging in designing further animal experiments that are lengthy and costly. Project 2 has the ultimate goal of identifying the genes and pathways associated with susceptibility to cancer risk from high LET radiation. The strategies employed are based on previous experience in using systems genetics approaches to identify germ line polymorphisms and somatic genetic changes associated with tumor development induced by chemicals or ionizing (low LET) radiation. This project is therefore focused heavily on genetic analysis of radiation-induced tumors, and as such complements the goals of Project 1 on analysis of the effects of radiation

exposure on the normal or tumor microenvironment.

The tumor suppressor Trp53, which is regularly mutated in cancers, is often referred to as the guardian of the genome due to the multiplicity of anti-proliferative effects it can initiate to maintain genomic integrity under genotoxic stress conditions. To investigate the impact of both low and high LET-radiation upon carcinogenesis we used the Trp53 deltaP mouse model created by our collaborator Geoff Wahl. Germline deletion of a small N-terminal proline-rich region of Trp53 in an in vivo mouse model (Trp53 deltaP) causes no spontaneous tumors. Interestingly, upon exposure of the Trp53 deltaP mouse to both low and high LET (linear energy transfer) radiation, a wide range of solid tissue tumors develops, including those of the liver, lung, and kidney. This is in stark contrast to the tumor spectrum in the Trp53 null (-/-) mouse post-radiation, which comprises mainly lymphomas and sarcomas. The Trp53 deltaP mouse therefore represents an exquisitely radiation-sensitive model to study radiation-induced tumors similar to those seen in the human population. Using a systems genetics approach as described above, we will generate an integrated network view of the factors which influence the susceptibility of certain tissues to the effects of both low and high LET radiation. Furthermore, this analysis will enable the identification of specific genes and pathways which are associated with the tumor-inducing effect of radiation. These data, in combination with next-generation sequencing data, enabling the identification of recurrent genetic aberrations occurring during tumorigenesis will provide insight to identifying potential genetic susceptibilities for future assessment in humans.

Task Progress:

In previous experiments we have observed a wide range of tumor types in various organs following irradiation of the Trp53 deltaP mouse on the original mixed background of 129/Sv/C57BL/6. Interestingly the irradiated tumor spectrum in the FVB/N background of both Trp53 deltaP and Trp53+/- mice falls into three main tumor types, thymic lymphomas, mammary tumors, and skin carcinomas. It should be noted that lymphomas are often observed post-radiation in mice of compromised Trp53 function, particularly after exposure to relatively high doses of ionizing radiation (4Gy). After exposure to lower doses (50cGy) of either low or high LET radiation, we observed an increased frequency of mammary tumors and skin carcinomas also in Trp53+/- mice. Although the tumor spectrum is similar between mice exposed to high or low LET radiation, there is an increased incidence of mammary tumors following high LET radiation.

Of note the tumor spectrum in 129/Sv Trp53+/- mice was markedly different from FVB/N. Regardless of radiation quality there was a high incidence of thymic lymphomas, and a wide range of tumors of varying types with no distinguishable pattern and also no skin carcinomas were observed. A very small subset of mammary tumors developed following high LET radiation (n=3), whilst none were observed in the gamma irradiated group. These differences further illustrate the influence of background strain upon tumor latency, susceptibility and spectrum following exposure to high or low LET radiation. These data demonstrate that FVB/N mice are particularly susceptible to radiation-induced mammary tumor development. High LET radiation is also more potent at inducing a higher incidence of mammary tumors, and shorter latency compared to gamma radiation (p=0.0004 CoxPH).

Little is known about the mechanisms by which high LET induces carcinogenesis. Exposing mice of abrogated Trp53 function to either high or low LET radiation provides us with a unique opportunity to compare and contrast the genetic aberrations and potential radiation-quality specific signatures in tumors. Next generation sequencing offers a powerful tool to gain a deeper understanding of the complex mutational processes driving radiation-induced tumorigenesis. In collaboration with Dr. David Adams at Sanger Institute, Cambridge, exome sequencing has been performed on 60 tumor samples, along with 10 control samples. We sequenced the most frequently observed tumor types, (thymic lymphomas, mammary tumors and skin carcinomas), in Trp53 deltaP and Trp53+/- on FVB/N or 129/Sv/C57BL/6 backgrounds induced by either Fe-ion or gamma radiation.

Project 3

Predicting cell survival RBE for any HZE using microdosimetry and a DNA damage clustering model. A manuscript has been submitted that predicts RBE for cell death based on experimental measurements done with X-ray. Briefly, Linear-quadratic models have been used for decades to interpret cell survival after exposure to ionizing radiation. However, parameters for these models change with LET, forcing investigators to measure experimentally cell survival for any LET of interest. In contrast to the current paradigm, we hypothesize that DSBs are moved into repair domains and consequently can be merged into clusters. Experimental work characterizing the dose dependence of radiation-induced foci (RIF) from X-ray in nonmalignant human mammary epithelial cells (MCF10A) is used here to validate a DSB clustering model. Cell death probability is computed as a decreasing exponential proportional to the number of DSBs and all combinations of pairs of DSBs that can occur in each cluster. This leads to two cell death coefficients from all DSBs and paired DSBs respectively. These coefficients can be inferred via computer simulations, by fitting survival curves of MCF10A exposed to X-ray. The clustering model that led to the best predictions assumes that the nucleus is divided into an array of regularly spaced repair domains. This model was then tested to predict the dose response of RIF and cell death after exposure to densely ionizing particles, as opposed to X-ray. Predicted RIF/μm had an average relative error of 12% when compared to experimental values covering a range of LET between 30 and 350 keV/μm for three different ions. Cell survival RBE predictions for MCF10A exposed to any LET remain to be verified experimentally. However, the predicted values and the LET dependence we report here match previous experimental results on similar cell types. With such an approach, parameters derived experimentally on X-ray are sufficient to predict the relative biological effectiveness of high-LET radiation by simply including microdosimetric properties of ion tracks. Overall, this work suggests that microdosimetric properties of ion tracks at the sub-micron level are sufficient to explain both RIF data and survival curves for any LET, similarly to the Local Effect Model (LEM) assumption. On the other hand, high-LET death mechanism does not have to infer linear-quadratic dose formalism as done in the LEM. Our model is therefore an alternative to previous approaches by providing a testable biological mechanism (i.e. RIF).This approach opens the door to a unified model to predict risk to cosmic rays for astronauts and we plan to extend it to also model RBE prediction for genomic instability.

Current development: integrating multi-stage clonal expansion model with NTE agent-based approaches. Our goal for the grant is to simulate tumor data in silico by assuming that tumors rises via successive DNA mutation and that promotion are influenced by non-targeted effects. One challenge in generating tumor in silico is the duration of simulations using our current ABM. We have switched to a simpler computer model (automata) that allows cells to be tracked faster over a 60 year simulation (from age 20 to 80 years old). We have currently implemented the multi-stage clonal expansion model (MSCE) to simulate tumor incidence arising spontaneously in human population due to random mutations using these automata. Each simulated person is a monolayer of 4 million cells (referred as “tissue”), where cells divide only when neighboring cells have died. Parameter sweep was done so that simulations predicted cancer incidence matching normal human population (~50% depending on sources). Our current parameters values are as

	follow: cell death frequency (beta term in MSCE) was set to 0.008, mutation frequency was set to 3E-8 per division, cell cycle time is set to 24 hours (experimental value) and we assume tumor arises via two events: initiation and conversion (v and μ term respectively in Two-Stage CE). The model is set to allow more stages before tumor arises if necessary (i.e. MSCE). Once a cell has become fully transformed, it can invade neighboring cells with an adjustable parameter for invasion efficiency. We plan to include radiation effects at two levels: targeted effect (as described in previous section) generating transient increase of mutation rate and non-targeted effects by modifying transformation rates transiently (transient time matching measurement made in this work and others).
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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